

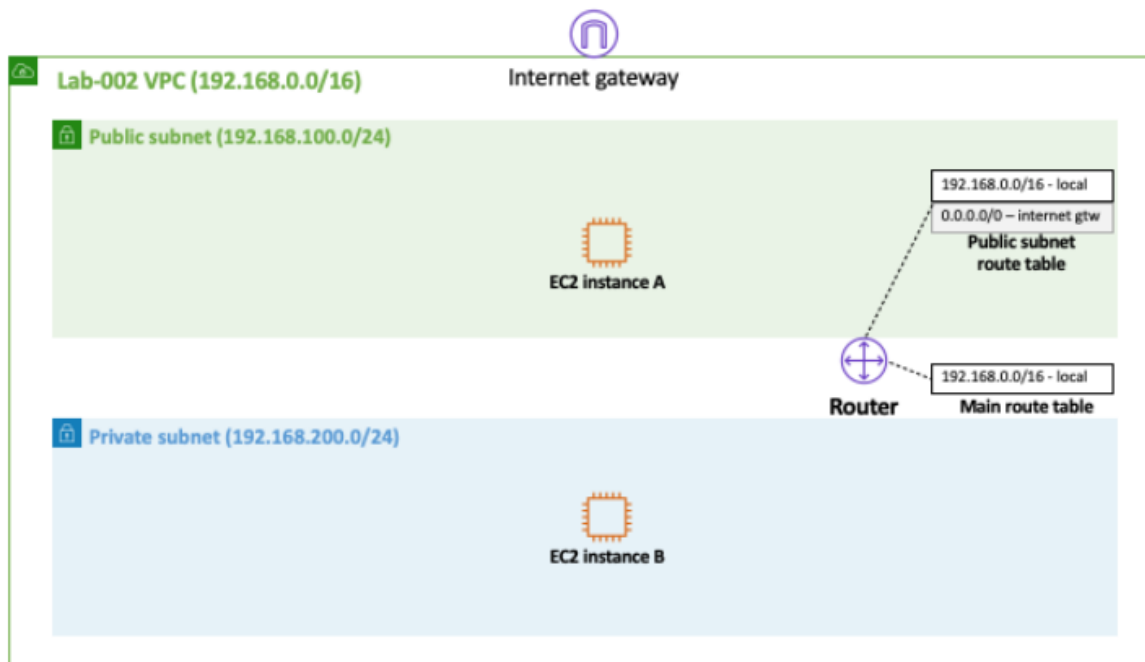
Lab-002

A Single EC2 Instance in a Private Subnet + Bastion Host

Goal

The goal of this lab is to demonstrate how to access an EC2 instance launched in a private subnet using a bastion host.

Architecture Diagram



Overview

In order to achieve the goal of this lab, you will have to go through the following steps:

Step 1 - Create VPC

Create a new VPC with the *Name tag* lab-002 and the *IPv4 CIDR block* 192.168.0.0/16.

[VPCs](#) > Create VPC

Create VPC

A VPC is an isolated portion of the AWS cloud populated by AWS objects, such as Amazon EC2 instances. You must specify an IPv4 address range for your VPC. Specify the IPv4 address range as a Classless Inter-Domain Routing (CIDR) block; for example, 10.0.0.0/16. You cannot specify an IPv4 CIDR block larger than /16. You can optionally associate an IPv6 CIDR block with the VPC.

Name tag ⓘ

IPv4 CIDR block* ⓘ

IPv6 CIDR block ☒ No IPv6 CIDR Block ⓘ
☐ Amazon provided IPv6 CIDR block

Tenancy ⓘ

* Required

[Cancel](#)

[Create](#)

[VPCs](#) > Create VPC

Create VPC

✓ The following VPC was created:

VPC ID [vpc-0344b484db0892dc6](#)

[Close](#)

Step 2 - Create an Internet Gateway

Create an [Internet gateway](#) with the *Name tag* lab-002 and attach it to lab-002 VPC.

Internet gateways (1/1) [Info](#)

Actions ▾

Create Internet gateway

< 1 >



☒	Name ▾	Internet gateway ID ▾	State ▾	VPC ID
☒	-	igw-0bb0f6fd2bc46d356	✔ Attached	vpc-924e57f

Create internet gateway [Info](#)

An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.

Internet gateway settings

Name tag

Creates a tag with a key of 'Name' and a value that you specify.

Cancel

Create Internet gateway

igw-0c3b9e827c57c370e / lab-002

Actions ▲

Attach to VPC

Detach from VPC

Manage tags

Delete

Details [Info](#)

Internet gateway ID	State	VPC ID	Owner ID
igw-0c3b9e827c57c370e	Detached	-	939096940193

VPC > Internet gateways > Attach to VPC (igw-0c3b9e827c57c370e)

Attach to VPC (igw-0c3b9e827c57c370e) Info

VPC

Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.

Available VPCs

Make sure to select the VPC with the name tag "lab-002"

Attach the internet gateway to this VPC

Q vpc-0344b484db0892dc6 X

► AWS Command Line Interface command

Cancel

Attach Internet gateway

VPC > Internet gateways

Internet gateways (2) Info

↺

Actions ▼

Create Internet gateway

Q Filter internet gateways

< 1 >

⚙

☐	Name ▼	Internet gateway ID ▼	State ▼	VPC ID
☐	-	igw-0bb0f6fd2bc46d356	✔ Attached	vpc-924e57f
☐	lab-002	igw-0c3b9e827c57c370e	✔ Attached	vpc-0344b48

Step 3 - Create the Public Subnet

Create a public subnet on lab-002 VPC with the *Name tag* public and *IPv4 CIDR block* 192.168.100.0/24.

Create subnet

Actions

Filter by tags and attributes or search by keyword

☐	Name	Subnet ID	State	VPC
☐	private	subnet-63b57d05	available	vpc-924e57f5 default
☐	public	subnet-b1b256eb	available	vpc-924e57f5 default

Subnets > Create subnet

Create subnet

Specify your subnet's IP address block in CIDR format; for example, 10.0.0.0/24. IPv4 block sizes must be between a /16 netmask and /28 netmask, and can be the same size as your VPC. An IPv6 CIDR block must be a /64 CIDR block.

Name tag

public

VPC*

vpc-0344b484db0892dc6

Availability Zone

Filter by attributes

vpc-0344b484db0892dc6lab-002

VPC CIDRs

vpc-924e57f5default

192.168.0.0/16associated

IPv4 CIDR block*

192.168.100.0/24

* Required

Cancel

Create

Subnets > Create subnet

Create subnet

✓

The following Subnet was created:

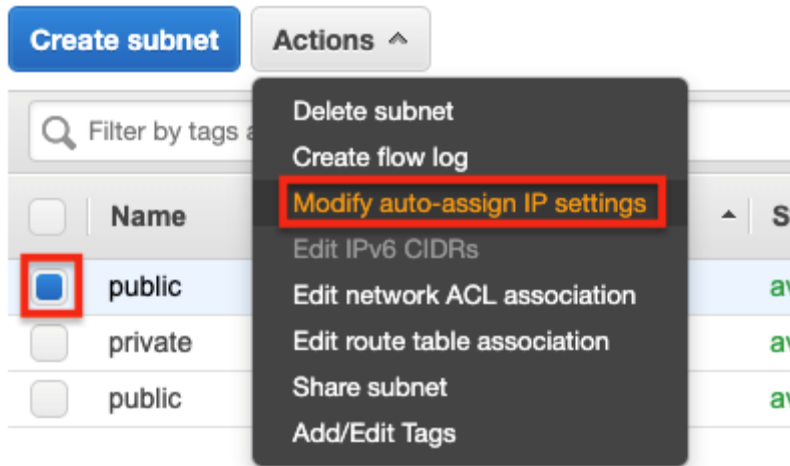
Subnet ID

subnet-08badf1a08badf598

Close

Step 4 - Enable Auto-assign IPv4

Enable *Auto-assign IPv4* in the newly created public subnet.



[Subnets](#) > Modify auto-assign IP settings

Modify auto-assign IP settings

Enable the auto-assign IP address setting to automatically request a public IPv4 or IPv6 address for an instance launched in this subnet. You can override the auto-assign IP settings for an instance at launch time.

Subnet ID subnet-08badf1a08badf598

Auto-assign IPv4 ☒ Enable auto-assign public IPv4 address ⓘ

* Required

[Cancel](#)

[Save](#)

Step 5 - Create a Route Table

Create a new route table with the *Name tag* public and with a default route to the internet gateway created in step 2.

Create route table

Actions

Filter by tags and attributes or search by keyword

<< 1 to 2 of 2 >>

☐	Name	Route Table ID	Explicit subnet association	Edge associations	Main	VPC ID	Owner
☒		rtb-03cb78a03f95efc35	-	-	Yes	vpc-0d8181c524284e897 ...	939096940
☐		rtb-a31287c5	-	-	Yes	vpc-924e57f5 default	939096940

Route Table: rtb-03cb78a03f95efc35

Summary

Routes

Subnet Associations

Edge Associations

Route Propagation

Tags

Route Table ID

rtb-03cb78a03f95efc35

Main

Yes

Explicitly Associated with

-

Owner

939096940193

VPC

vpc-0d8181c524284e897 | lab-002

Create route table

Actions

Filter by tags and attributes or search by keyword

<< 1 to 2 of 2 >>

☐	Name	Route Table ID	Explicit subnet association	Edge associations	Main	VPC ID	Owner
☒	main	rtb-03cb78a03f95efc35	-	-	Yes	vpc-0d8181c524284e897 ...	939096940
☐		rtb-a31287c5	-	-	Yes	vpc-924e57f5 default	939096940

Route Table: rtb-03cb78a03f95efc35

Summary

Routes

Subnet Associations

Edge Associations

Route Propagation

Tags

Route Table ID

rtb-03cb78a03f95efc35

Main

Yes

Explicitly Associated with

-

Owner

939096940193

VPC

vpc-0d8181c524284e897 | lab-002

Route Tables > Create route table

Create route table

A route table specifies how packets are forwarded between the subnets within your VPC, the Internet, and your VPN connection.

Name tag

public

VPC*

vpc-0d8181c524284e897

* Required

Filter by attributes

vpc-924e57f5 default

vpc-0d8181c524284e897 lab-002

Cancel

Create

Route Tables > Create route table

Create route table

✔ The following Route Table was created:

Route Table ID

rtb-0d297ed554058cbb5

Close

Create route tableActions

Filter by tags and attributes or search by keyword

<>1 to 3 of 3>>

☐	Name	Route Table ID	Explicit subnet association	Edge associations	Main	VPC ID	Owner
☐	main	rtb-03cb78a03f95efc35	-	-	Yes	vpc-0d8181c524284e897 ...	939096940
☒	public	rtb-0d297ed554058cbb5	-	-	No	vpc-0d8181c524284e897 ...	939096940
☐		rtb-a31287c5	-	-	Yes	vpc-924e57f5 default	939096940

Route Table: rtb-0d297ed554058cbb5

Summary

Routes

Subnet Associations

Edge Associations

Route Propagation

Tags

Edit routes

ViewAll routes

Destination	Target	Status	Propagated
192.168.0.0/16	local	active	No

Route Tables > Edit routes

Edit routes

Destination	Target	Status	Propagated
192.168.0.0/16	local	active	No

Add route

* Required

CancelSave routes

Route Tables > Edit routes

Edit routes

Destination	Target	Status	Propagated
192.168.0.0/16	local	active	No
0.0.0.0/0	igw-		No

Add route

igw-0a35b250bdf127629lab-002

* Required

CancelSave routes

Route Tables > Edit routes

Edit routes

Routes successfully edited

Close

Step 6 - Associate the Route Table to the Public Subnet

Associate the newly created route table to the public subnet.

Create route table

Actions

Filter by tags and attributes or search by keyword

☐	Name	Route Table ID	Explicit subnet association	Edge associations
☐	main	rtb-03cb78a03f95efc35	-	-
☒	public	rtb-0d297ed554058cbb5	-	-
☐		rtb-a31287c5	-	-

Route Table: rtb-0d297ed554058cbb5

Summary

Routes

Subnet Associations

Edge Associations

Route Propagation

Edit subnet associations

Subnet ID	IPv4 CIDR	IPv6 CIDR
-----------	-----------	-----------

You do not have any subnet associations.

[Route Tables](#) > Edit subnet associations

Edit subnet associations

Route table rtb-0d297ed554058cbb5 (public)

Associated subnets subnet-0b9f50af8cac77468

Filter by attributes or search by keyword

< 1 to 1 of 1 >

☐	Subnet ID	IPv4 CIDR	IPv6 CIDR	Current Route Table
☒	subnet-0b9f50af8cac77468 public	192.168.100....	-	Main

* Required

[Cancel](#) [Save](#)

Step 7 - Create the Private Subnet

Create a private subnet on lab-002 VPC with the *Name tag* private and the *IPv4 CIDR block* 192.168.200.0/24.

Create subnet

Actions

Filter by tags and attributes or search by keyword

☐	Name	Subnet ID	State	VPC	IP
☐	public	subnet-08badf1a08badf598	available	vpc-0344b484db0892dc6 ...	19
☐	private	subnet-63b57d05	available	vpc-924e57f5 default	17
☐	public	subnet-b1b256eb	available	vpc-924e57f5 default	17

Subnets > Create subnet

Create subnet

Specify your subnet's IP address block in CIDR format; for example, 10.0.0.0/24. IPv4 block sizes must be between a /16 netmask and /28 netmask, and can be the same size as your VPC. An IPv6 CIDR block must be a /64 CIDR block.

Name tag

private

VPC*

Availability Zone

Filter by attributes

vpc-0344b484db0892dc6

lab-002

VPC CIDRs

vpc-924e57f5

default

Status Reason

IPv4 CIDR block*

192.168.200.0/24

* Required

Cancel

Create

Subnets > Create subnet

Create subnet

✔ The following Subnet was created:

Subnet ID subnet-03f060ca96dca8186

Close

Step 8 - Launch the EC2 Instances

Launch two EC2 instances using the procedure described in [lab-001](#); make sure that each subnet has one of the instances.

Note that the EC2 instance that you will create in the public subnet (labeled as A in the diagram) is necessary so you can later connect to the EC2 instance in the private subnet (labeled as B in the diagram). EC2 A is normally called *bastion host* or *jump host*.

Test and Validation

1. First add the EC2 key pair into your local ssh authentication agent using:

```
ssh-add -K lab-002.pem (macos)
ssh-add -c lab-002.pem (linux)
```

2. Then connect to the EC2 instance A but with ssh agent forwarding enabled:

```
ssh -A ec2-user@<public-IP of A>
```

3. Finally, from EC2 instance A ssh to the EC2 instance B using its private IP address:

```
ssh <private-IP of B>
```